

A Complete Integer Solution of

$$y^2 + x^3y + z^2 + 1 = 0$$

Abstract

We solve the Diophantine equation

$$y^2 + x^3y + z^2 + 1 = 0$$

completely over the integers. The solution is reduced, by completing the square, to representations of $x^6 - 4$ as a sum of two squares. For completeness, the classical two-squares criterion is proved from first principles using the Euclidean arithmetic of the Gaussian integers.

1 Statement of the answer

For a positive integer n , define

$$\mathcal{R}_n = \{(a, b) \in \mathbb{Z}_{>0}^2 : a^2 + b^2 = n^6 - 4, \ a \equiv 1 \pmod{2}, \ b \equiv 0 \pmod{2}\}.$$

Also define

$$\mathcal{N} = \{n \in \mathbb{Z}_{>0} : n \equiv 3 \pmod{12} \text{ and } v_p(n^6 - 4) \text{ is even for every prime } p \equiv 3 \pmod{4}\}.$$

Theorem 1 (Complete solution). *The integer solutions of*

$$y^2 + x^3y + z^2 + 1 = 0$$

are exactly the triples

$$\boxed{x = \varepsilon n, \quad y = \frac{\sigma a - \varepsilon n^3}{2}, \quad z = \tau \frac{b}{2}}$$

where

$$n \in \mathcal{N}, \quad (a, b) \in \mathcal{R}_n, \quad \varepsilon, \sigma, \tau \in \{\pm 1\}.$$

Equivalently, for a positive integer n , one has $\mathcal{R}_n \neq \emptyset$ if and only if

$$n \equiv 3 \pmod{12}$$

and every prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in $n^6 - 4$.

The proof occupies the rest of the note. The formulation above is not a finite table of triples; it is a complete parametrization together with an effective criterion. For a fixed n , one factors $n^6 - 4$, checks the exponents of the primes congruent to 3 modulo 4, and then computes the finitely many sum-of-two-squares representations of $n^6 - 4$.

2 Elementary reduction to a sum of two squares

Lemma 2 (The square-completion identity). *For integers x, y, z , the equation*

$$y^2 + x^3y + z^2 + 1 = 0$$

is equivalent to

$$(2y + x^3)^2 + (2z)^2 = x^6 - 4.$$

Proof. Multiplying the original equation by 4 gives

$$4y^2 + 4x^3y + 4z^2 + 4 = 0.$$

Since

$$4y^2 + 4x^3y = (2y + x^3)^2 - x^6,$$

this is equivalent to

$$(2y + x^3)^2 - x^6 + 4z^2 + 4 = 0,$$

which is precisely

$$(2y + x^3)^2 + (2z)^2 = x^6 - 4.$$

The algebra is reversible, so the two equations are equivalent. □

Lemma 3 (First congruence obstructions). *If $(x, y, z) \in \mathbb{Z}^3$ satisfies*

$$y^2 + x^3y + z^2 + 1 = 0,$$

then x is odd and $3 \mid x$.

Proof. First suppose that x is even. Then x^3y is divisible by 4, and the original equation gives

$$y^2 + z^2 + 1 \equiv 0 \pmod{4}.$$

But a square modulo 4 is either 0 or 1, so $y^2 + z^2$ is congruent only to 0, 1, or 2 modulo 4, never to 3 modulo 4. This contradiction proves that x is odd.

Now suppose that $3 \nmid x$. By the square-completion identity,

$$(2y + x^3)^2 + (2z)^2 = x^6 - 4.$$

The quadratic residues modulo 9 are

$$0, 1, 4, 7.$$

Indeed, this is obtained by squaring the residue classes $0, 1, \dots, 8$. No sum of two elements of $\{0, 1, 4, 7\}$ is congruent to 6 modulo 9: the possible sums are

$$0, 1, 2, 4, 5, 7, 8 \pmod{9}.$$

On the other hand, if $3 \nmid x$, then $x^6 \equiv 1 \pmod{9}$, and hence

$$x^6 - 4 \equiv 1 - 4 \equiv 6 \pmod{9}.$$

Thus the right-hand side of the square-completion identity is congruent to 6 modulo 9, while the left-hand side is a sum of two squares modulo 9, impossible. Therefore $3 \mid x$. □

Lemma 4 (Parity of representations). *Let n be odd and suppose*

$$a^2 + b^2 = n^6 - 4$$

with $a, b \in \mathbb{Z}$. Then neither a nor b is zero. Moreover, exactly one of a, b is odd and the other is even. After interchanging the two coordinates if necessary, the representation may be written with a odd and b even.

Proof. Since n is odd, $n^2 \equiv 1 \pmod{8}$, so $n^6 \equiv 1 \pmod{8}$. Hence

$$n^6 - 4 \equiv 5 \pmod{8}.$$

No square is congruent to 5 modulo 8, so neither coordinate can be zero. Also, because $n^6 - 4$ is odd, exactly one of a, b is odd and the other is even. Therefore we can label the odd coordinate by a and the even coordinate by b . $\square$

3 The two-squares theorem, proved self-containedly

We need the classical criterion for an integer to be a sum of two squares. To keep the solution self-contained, we prove it here.

3.1 Gaussian integers

Definition 5. The Gaussian integers are

$$\mathbb{Z}[i] = \{u + vi : u, v \in \mathbb{Z}\}.$$

For $\alpha = u + vi \in \mathbb{Z}[i]$, define its norm by

$$N(\alpha) = \alpha\bar{\alpha} = u^2 + v^2.$$

Thus $N(\alpha)$ is a nonnegative rational integer, and $N(\alpha) = 0$ if and only if $\alpha = 0$.

Definition 6 (Divisibility terminology). For $\alpha, \beta \in \mathbb{Z}[i]$, we write $\alpha \mid \beta$ if $\beta = \alpha\gamma$ for some $\gamma \in \mathbb{Z}[i]$. A Gaussian integer u is a unit if it divides 1. Two Gaussian integers are associates if one is obtained from the other by multiplication by a unit. A nonzero nonunit $\pi \in \mathbb{Z}[i]$ is irreducible if every factorization $\pi = \alpha\beta$ has one of α, β a unit. A nonzero nonunit $\pi \in \mathbb{Z}[i]$ is prime if

$$\pi \mid \alpha\beta \implies \pi \mid \alpha \text{ or } \pi \mid \beta$$

for all $\alpha, \beta \in \mathbb{Z}[i]$.

Lemma 7 (Units). *The units of $\mathbb{Z}[i]$ are precisely*

$$\pm 1, \pm i.$$

Equivalently, $\alpha \in \mathbb{Z}[i]$ is a unit if and only if $N(\alpha) = 1$.

Proof. If α is a unit, then $\alpha\beta = 1$ for some $\beta \in \mathbb{Z}[i]$. Taking norms gives

$$N(\alpha)N(\beta) = 1.$$

Both factors are nonnegative integers, so $N(\alpha) = 1$. Writing $\alpha = u + vi$, this means

$$u^2 + v^2 = 1,$$

so $(u, v) = (\pm 1, 0)$ or $(0, \pm 1)$, giving $\alpha = \pm 1$ or $\alpha = \pm i$. Conversely, each of $\pm 1, \pm i$ plainly has a multiplicative inverse in $\mathbb{Z}[i]$. $\square$

Lemma 8 (Multiplicativity of the norm). *For all $\alpha, \beta \in \mathbb{Z}[i]$,*

$$N(\alpha\beta) = N(\alpha) N(\beta).$$

Proof. Using complex conjugation,

$$N(\alpha\beta) = \alpha\beta\overline{\alpha\beta} = \alpha\beta\overline{\alpha}\overline{\beta} = (\alpha\overline{\alpha})(\beta\overline{\beta}) = N(\alpha) N(\beta).$$

□

Lemma 9 (Euclidean division in $\mathbb{Z}[i]$). *For any $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$, there exist $q, r \in \mathbb{Z}[i]$ such that*

$$\alpha = q\beta + r \quad \text{and} \quad N(r) < N(\beta).$$

Proof. Write

$$\frac{\alpha}{\beta} = s + ti$$

with $s, t \in \mathbb{R}$. Choose integers m, n satisfying

$$|s - m| \leq \frac{1}{2}, \quad |t - n| \leq \frac{1}{2},$$

and set $q = m + ni \in \mathbb{Z}[i]$. Define $r = \alpha - q\beta$. Then

$$r = \beta \left(\frac{\alpha}{\beta} - q \right),$$

and therefore, using multiplicativity of the norm and the fact that the norm agrees with the squared complex modulus,

$$N(r) = N(\beta) \left| \frac{\alpha}{\beta} - q \right|^2 \leq N(\beta) \left(\frac{1}{4} + \frac{1}{4} \right) = \frac{1}{2} N(\beta) < N(\beta).$$

Thus the required division with remainder exists. □

Lemma 10 (Greatest common divisors and Bezout identity in $\mathbb{Z}[i]$). *Let $\alpha, \beta \in \mathbb{Z}[i]$ be not both zero. Then there is a Gaussian integer d such that*

$$d \mid \alpha, \quad d \mid \beta,$$

and every common divisor of α and β divides d . Moreover, there exist $u, v \in \mathbb{Z}[i]$ such that

$$d = u\alpha + v\beta.$$

Such a d is unique up to multiplication by a unit.

Proof. If one of α, β is zero, the assertion is immediate: for example, if $\beta = 0$, take $d = \alpha$, $u = 1$, and $v = 0$.

Assume now that $\alpha\beta \neq 0$. Apply Euclidean division repeatedly. Set

$$r_{-1} = \alpha, \quad r_0 = \beta.$$

If $r_j \neq 0$, divide r_{j-1} by r_j to obtain

$$r_{j-1} = q_j r_j + r_{j+1}, \quad N(r_{j+1}) < N(r_j),$$

unless $r_{j+1} = 0$, in which case the algorithm stops. Since the norms

$$N(r_0) > N(r_1) > N(r_2) > \cdots$$

are nonnegative integers whenever the corresponding remainders are nonzero, the process terminates. Let r_m be the last nonzero remainder.

The last division gives $r_{m-1} = q_m r_m$, so $r_m \mid r_{m-1}$. The recurrence

$$r_{j-1} = q_j r_j + r_{j+1}$$

then shows, by backward induction, that r_m divides every preceding remainder, in particular $r_m \mid \alpha$ and $r_m \mid \beta$.

Conversely, let c be any common divisor of α and β . From

$$r_{j+1} = r_{j-1} - q_j r_j$$

it follows by forward induction that c divides every remainder r_j . Hence $c \mid r_m$. Thus r_m is a greatest common divisor in the divisibility sense.

Finally, each remainder is a $\mathbb{Z}[i]$ -linear combination of α and β : this is true for r_{-1} and r_0 , and the recurrence

$$r_{j+1} = r_{j-1} - q_j r_j$$

preserves the property. Therefore $r_m = u\alpha + v\beta$ for some $u, v \in \mathbb{Z}[i]$. Put $d = r_m$.

If d and d' are two greatest common divisors, then $d \mid d'$ and $d' \mid d$. Thus $d' = d\eta$ and $d = d'\theta$ for some $\eta, \theta \in \mathbb{Z}[i]$, so $d = d\eta\theta$. Since $d \neq 0$, cancellation in the integral domain $\mathbb{Z}[i]$ gives $\eta\theta = 1$, and η is a unit. Hence d and d' are associates. $\square$

Lemma 11 (Irreducibles are prime in $\mathbb{Z}[i]$). *Every irreducible Gaussian integer is prime.*

Proof. Let $\pi \in \mathbb{Z}[i]$ be irreducible and suppose

$$\pi \mid \alpha\beta.$$

Let d be a greatest common divisor of π and α . Since $d \mid \pi$ and π is irreducible, either d is a unit or d is associated to π .

If d is associated to π , then $\pi \mid \alpha$. If d is a unit, Bezout's identity gives Gaussian integers u, v such that

$$u\pi + v\alpha = 1.$$

Multiplying by β gives

$$u\pi\beta + v\alpha\beta = \beta.$$

The first term on the left is divisible by π , and the second term is divisible by π because $\pi \mid \alpha\beta$. Therefore $\pi \mid \beta$. Hence π divides at least one of α, β , proving that π is prime. $\square$

Lemma 12 (Existence of irreducible factorizations in $\mathbb{Z}[i]$). *Every nonzero nonunit Gaussian integer factors as a finite product of irreducible Gaussian integers.*

Proof. Suppose not. Among all nonzero nonunits that do not admit a factorization into irreducibles, choose one, say γ , with $N(\gamma)$ minimal. The element γ cannot itself be irreducible. Hence

$$\gamma = \delta\eta$$

with δ, η both nonunits. Since δ and η are nonzero nonunits, their norms are integers greater than 1. By multiplicativity,

$$N(\gamma) = N(\delta)N(\eta),$$

so

$$1 < N(\delta) < N(\gamma), \quad 1 < N(\eta) < N(\gamma).$$

By the minimal choice of γ , both δ and η factor into irreducibles. Multiplying those two factorizations gives a factorization of γ into irreducibles, a contradiction. $\square$

Lemma 13 ($\mathbb{Z}[i]$ is a unique factorization domain). *Every nonzero nonunit Gaussian integer factors as a product of irreducible Gaussian integers, and this factorization is unique up to multiplication by units and reordering.*

Proof. Existence is the preceding lemma. We prove uniqueness.

Suppose

$$\pi_1 \pi_2 \cdots \pi_r = \rho_1 \rho_2 \cdots \rho_s$$

are two factorizations into irreducibles. Since π_1 is irreducible, it is prime by the preceding lemma. Therefore π_1 divides one of the factors on the right, say ρ_j . Because ρ_j is irreducible and π_1 is a nonunit divisor of ρ_j , the quotient ρ_j/π_1 is a unit; equivalently, π_1 and ρ_j are associates.

After reordering the ρ 's, write $\rho_1 = u\pi_1$ with u a unit. Cancelling π_1 from the equality gives

$$\pi_2 \cdots \pi_r = u\rho_2 \cdots \rho_s.$$

Absorb the unit u into one of the remaining factors if any remain; if no factor remains, the equality says that the remaining product on the other side is a unit, which is possible only when it is empty. Repeating the argument by induction on $r + s$ matches all irreducible factors up to associates. Therefore the factorization is unique up to multiplication by units and reordering. $\square$

3.2 Rational primes in $\mathbb{Z}[i]$

Lemma 14 (Fermat's little theorem). *Let p be a rational prime and let $u \in \mathbb{Z}$ with $p \nmid u$. Then*

$$u^{p-1} \equiv 1 \pmod{p}.$$

Proof. Multiplication by u permutes the nonzero residue classes modulo p . Therefore

$$u \cdot 2u \cdot 3u \cdots (p-1)u \equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \pmod{p}.$$

The right-hand product is not divisible by p , so it may be cancelled modulo p , giving

$$u^{p-1} \equiv 1 \pmod{p}.$$

$\square$

Lemma 15 (Wilson's theorem). *If p is a rational prime, then*

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. If $p = 2$, then $(p - 1)! = 1$ and $-1 \equiv 1 \pmod{2}$, so the assertion is true.

Assume now that p is odd. Modulo p , each nonzero residue has a multiplicative inverse. The residues equal to their own inverses are exactly the solutions of

$$t^2 \equiv 1 \pmod{p},$$

or equivalently

$$(t - 1)(t + 1) \equiv 0 \pmod{p}.$$

Since p is prime, this gives precisely

$$t \equiv 1 \pmod{p} \quad \text{or} \quad t \equiv -1 \pmod{p}.$$

Because p is odd, these two residues are distinct. Therefore, in the product of all nonzero residues modulo p , every residue other than 1 and -1 pairs with its distinct inverse, and each such pair contributes 1. Hence

$$(p - 1)! \equiv 1 \cdot (-1) \equiv -1 \pmod{p}.$$

This proves Wilson's theorem for all primes p . □

Lemma 16 (The residue of -1 modulo an odd prime). *Let p be an odd rational prime.*

1. *If $p \equiv 3 \pmod{4}$, then -1 is not a quadratic residue modulo p .*
2. *If $p \equiv 1 \pmod{4}$, then -1 is a quadratic residue modulo p .*

Proof. First let $p \equiv 3 \pmod{4}$. If $t^2 \equiv -1 \pmod{p}$, then $p \nmid t$, so Fermat's little theorem gives $t^{p-1} \equiv 1 \pmod{p}$. But $(p - 1)/2$ is odd, and therefore

$$t^{p-1} = (t^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} \equiv -1 \pmod{p},$$

a contradiction. Hence -1 is not a quadratic residue modulo p .

Now let $p \equiv 1 \pmod{4}$. Wilson's theorem gives

$$(p - 1)! \equiv -1 \pmod{p}.$$

Pairing the factors j and $p - j$ for $1 \leq j \leq (p - 1)/2$ gives

$$(p - 1)! = \prod_{j=1}^{(p-1)/2} j(p - j) \equiv (-1)^{(p-1)/2} \left(\left(\frac{p-1}{2} \right)! \right)^2 \pmod{p}.$$

Since $p \equiv 1 \pmod{4}$, the exponent $(p - 1)/2$ is even. Therefore

$$\left(\left(\frac{p-1}{2} \right)! \right)^2 \equiv -1 \pmod{p},$$

so -1 is a quadratic residue modulo p . □

Lemma 17 (Behavior of rational primes in $\mathbb{Z}[i]$). *Let p be a rational prime.*

1. $2 = (1 + i)(1 - i)$, so 2 is a norm: $2 = N(1 + i)$.
2. *If $p \equiv 3 \pmod{4}$, then p is irreducible, hence prime, in $\mathbb{Z}[i]$.*

3. If $p \equiv 1 \pmod{4}$, then there are integers a, b such that

$$p = a^2 + b^2 = N(a + bi).$$

Equivalently, p factors nontrivially in $\mathbb{Z}[i]$.

Proof. The identity for 2 is immediate:

$$(1 + i)(1 - i) = 2.$$

Let $p \equiv 3 \pmod{4}$. Suppose p were reducible in $\mathbb{Z}[i]$. Then

$$p = \alpha\beta$$

with α, β nonunits. Taking norms gives

$$p^2 = N(\alpha)N(\beta).$$

Because α and β are nonunits, their norms exceed 1. Since p is rationally prime, this forces

$$N(\alpha) = N(\beta) = p.$$

Writing $\alpha = a + bi$, we obtain

$$a^2 + b^2 = p.$$

If $p \mid b$, then reducing $a^2 + b^2 = p$ modulo p gives $p \mid a$, whence $p^2 \mid a^2 + b^2 = p$, impossible. Thus b is invertible modulo p . Reducing $a^2 + b^2 = p$ modulo p gives

$$a^2 \equiv -b^2 \pmod{p},$$

and therefore

$$(ab^{-1})^2 \equiv -1 \pmod{p},$$

contradicting the fact that -1 is not a quadratic residue modulo primes $p \equiv 3 \pmod{4}$. Therefore p is irreducible in $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a unique factorization domain, irreducible elements are prime, so p is prime in $\mathbb{Z}[i]$.

Now let $p \equiv 1 \pmod{4}$. By the preceding lemma, choose $t \in \mathbb{Z}$ such that

$$t^2 \equiv -1 \pmod{p}.$$

Then

$$p \mid t^2 + 1 = (t + i)(t - i)$$

in $\mathbb{Z}[i]$. If p were prime in $\mathbb{Z}[i]$, it would divide $t + i$ or $t - i$. But $p \mid t + i$ in $\mathbb{Z}[i]$ would mean that both real and imaginary parts of $t + i$ are divisible by p , i.e. $p \mid t$ and $p \mid 1$, impossible. Similarly, $p \nmid t - i$. Hence p is not prime in $\mathbb{Z}[i]$, and therefore not irreducible. So

$$p = \alpha\beta$$

with α, β nonunits. Taking norms again gives

$$p^2 = N(\alpha)N(\beta),$$

and hence $N(\alpha) = p$ and $N(\beta) = p$. Writing $\alpha = a + bi$ gives

$$p = a^2 + b^2.$$

This proves the claim. □

3.3 The criterion

Theorem 18 (Two-squares criterion). *Let M be a positive integer. Then M is representable as*

$$M = A^2 + B^2$$

with $A, B \in \mathbb{Z}$ if and only if every rational prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in the prime factorization of M .

Proof. First suppose $M = A^2 + B^2$. Put

$$\alpha = A + Bi \in \mathbb{Z}[i].$$

Then

$$M = N(\alpha) = \alpha\bar{\alpha}.$$

Let $p \equiv 3 \pmod{4}$ be a rational prime. By the preceding lemma, p is prime in $\mathbb{Z}[i]$. In the unique factorization of α , suppose p occurs to exponent e . Since $\bar{p} = p$, the same exponent e occurs in the unique factorization of $\bar{\alpha}$. Therefore the exponent of p in

$$\alpha\bar{\alpha} = M$$

is $2e$, an even integer. Hence every rational prime $p \equiv 3 \pmod{4}$ has even exponent in M .

Conversely, suppose every prime $p \equiv 3 \pmod{4}$ has even exponent in M . Write the rational prime factorization of M as

$$M = 2^e \prod_{p \equiv 1 \pmod{4}} p^{e_p} \prod_{q \equiv 3 \pmod{4}} q^{2f_q},$$

where the products are finite and range only over primes that divide M . For 2, use $2 = N(1 + i)$. For each prime $p \equiv 1 \pmod{4}$, the preceding lemma gives a Gaussian integer π_p with

$$N(\pi_p) = p.$$

For each prime $q \equiv 3 \pmod{4}$, the Gaussian integer q itself has norm

$$N(q) = q^2.$$

Define

$$\gamma = (1 + i)^e \prod_{p \equiv 1 \pmod{4}} \pi_p^{e_p} \prod_{q \equiv 3 \pmod{4}} q^{f_q} \in \mathbb{Z}[i].$$

By multiplicativity of the norm,

$$N(\gamma) = M.$$

Writing $\gamma = A + Bi$ gives

$$M = A^2 + B^2.$$

This proves both directions. □

4 Completion of the Diophantine solution

We now return to

$$y^2 + x^3y + z^2 + 1 = 0.$$

Lemma 19 (Necessary form of $|x|$). *Let (x, y, z) be an integer solution, and put $n = |x|$. Then*

$$n \equiv 3 \pmod{12},$$

and every prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in $n^6 - 4$.

Proof. By the congruence lemma, x is odd and $3 \mid x$. Hence n is odd and $3 \mid n$. By the square-completion identity,

$$(2y + x^3)^2 + (2z)^2 = n^6 - 4.$$

Thus $n^6 - 4$ is a sum of two squares. By the two-squares criterion, every prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in $n^6 - 4$.

It remains to prove $n \equiv 3 \pmod{4}$. Since $3 \mid n$ and $n > 0$, we have $n \geq 3$. Suppose, for contradiction, that $n \equiv 1 \pmod{4}$. Then

$$n^3 - 2 \equiv 3 \pmod{4}, \quad n^3 + 2 \equiv 3 \pmod{4}.$$

Moreover,

$$\gcd(n^3 - 2, n^3 + 2) \mid 4.$$

Both $n^3 - 2$ and $n^3 + 2$ are odd, so in fact

$$\gcd(n^3 - 2, n^3 + 2) = 1.$$

Now an odd integer congruent to 3 modulo 4 must contain some prime congruent to 3 modulo 4 to an odd exponent: indeed, primes congruent to 1 modulo 4 contribute 1 modulo 4, while primes congruent to 3 modulo 4 contribute -1 modulo 4, so the total number of prime factors congruent to 3 modulo 4, counted with multiplicity, must be odd.

Therefore each of the coprime integers $n^3 - 2$ and $n^3 + 2$ contains a prime $\equiv 3 \pmod{4}$ to an odd exponent. Since the two factors are coprime, that prime remains to an odd exponent in their product

$$(n^3 - 2)(n^3 + 2) = n^6 - 4,$$

contradicting the two-squares criterion. Hence $n \not\equiv 1 \pmod{4}$. Since n is odd, $n \equiv 3 \pmod{4}$.

Combining $3 \mid n$ with $n \equiv 3 \pmod{4}$ gives

$$n \equiv 3 \pmod{12}.$$

This proves the lemma. □

Lemma 20 (Sufficiency and reconstruction). *Let $n \in \mathcal{N}$, let $(a, b) \in \mathcal{R}_n$, and choose signs*

$$\varepsilon, \sigma, \tau \in \{\pm 1\}.$$

Define

$$x = \varepsilon n, \quad y = \frac{\sigma a - \varepsilon n^3}{2}, \quad z = \tau \frac{b}{2}.$$

Then x, y, z are integers and

$$y^2 + x^3y + z^2 + 1 = 0.$$

Conversely, every integer solution arises in this way.

Proof. First observe that x is an integer. Since $n \equiv 3 \pmod{12}$, n is odd, so n^3 is odd. Since a is odd, $\sigma a - \varepsilon n^3$ is even; hence $y \in \mathbb{Z}$. Since b is even, $z \in \mathbb{Z}$.

Now compute

$$2y + x^3 = \sigma a - \varepsilon n^3 + (\varepsilon n)^3.$$

Because $\varepsilon^3 = \varepsilon$, this simplifies to

$$2y + x^3 = \sigma a.$$

Similarly,

$$2z = \tau b.$$

Therefore

$$(2y + x^3)^2 + (2z)^2 = a^2 + b^2 = n^6 - 4 = x^6 - 4.$$

By the square-completion identity, this is equivalent to

$$y^2 + x^3y + z^2 + 1 = 0.$$

Thus every triple constructed in the stated way is a solution.

Conversely, let (x, y, z) be any integer solution and set

$$n = |x|, \quad \varepsilon = \operatorname{sgn}(x).$$

The necessary-form lemma gives $n \in \mathcal{N}$. Define

$$A = 2y + x^3, \quad B = 2z.$$

Then

$$A^2 + B^2 = n^6 - 4.$$

Since x is odd, $A = 2y + x^3$ is odd, and $B = 2z$ is even. The parity lemma shows that neither A nor B is zero. Put

$$a = |A|, \quad b = |B|, \quad \sigma = \operatorname{sgn}(A), \quad \tau = \operatorname{sgn}(B).$$

Then $(a, b) \in \mathcal{R}_n$, and

$$2y + x^3 = A = \sigma a.$$

Since $x = \varepsilon n$, this gives

$$2y = \sigma a - \varepsilon n^3, \quad y = \frac{\sigma a - \varepsilon n^3}{2}.$$

Also

$$z = \tau \frac{b}{2}.$$

Thus the solution has exactly the asserted form. □

Lemma 21 (Non-emptiness of $\mathcal{R}_n$). *For a positive integer n , the set $\mathcal{R}_n$ is nonempty if and only if*

$$n \equiv 3 \pmod{12}$$

and every prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in $n^6 - 4$.

Proof. Suppose first that $\mathcal{R}_n$ is nonempty, and choose $(a, b) \in \mathcal{R}_n$. Then $a^2 + b^2 = n^6 - 4$, so the two-squares criterion gives the required even-exponent condition. Also, reducing $a^2 + b^2 = n^6 - 4$ modulo 2 gives $1 \equiv n^6 \pmod{2}$, so n is odd. Therefore

$$x = n, \quad y = \frac{a - n^3}{2}, \quad z = \frac{b}{2}$$

are integers, and the square-completion identity gives

$$y^2 + x^3y + z^2 + 1 = 0.$$

Applying the necessary-form lemma to this solution gives $n \equiv 3 \pmod{12}$.

Conversely, suppose $n \equiv 3 \pmod{12}$ and every prime $p \equiv 3 \pmod{4}$ occurs to an even exponent in $n^6 - 4$. By the two-squares criterion, there exist integers c, d such that

$$c^2 + d^2 = n^6 - 4.$$

Because n is odd,

$$n^6 - 4 \equiv 5 \pmod{8}.$$

The parity lemma then shows that neither c nor d is zero and exactly one of them is odd. Taking absolute values and, if necessary, interchanging the two coordinates, we obtain positive integers a, b with

$$a^2 + b^2 = n^6 - 4, \quad a \text{ odd}, \quad b \text{ even}.$$

Thus $(a, b) \in \mathcal{R}_n$, so $\mathcal{R}_n \neq \emptyset$. □

The complete theorem follows from the reconstruction lemma and the non-emptiness lemma.

5 Examples

For $n = 3$, one has

$$3^6 - 4 = 729 - 4 = 725.$$

The positive representations with the first coordinate odd and the second even are

$$725 = 25^2 + 10^2 = 23^2 + 14^2 = 7^2 + 26^2.$$

Taking $x = 3$ gives the following solutions:

(a, b)	y	z
$(25, 10)$	$-1, -26$	± 5
$(23, 14)$	$-2, -25$	± 7
$(7, 26)$	$-10, -17$	± 13

Thus, for example,

$$(x, y, z) = (3, -1, 5)$$

is a solution, since

$$(-1)^2 + 3^3(-1) + 5^2 + 1 = 1 - 27 + 25 + 1 = 0.$$

The symmetry

$$(x, y, z) \mapsto (-x, -y, z)$$

preserves the equation, so the corresponding solutions with $x = -3$ are obtained by changing the sign of y .

A larger example is obtained from

$$87^6 - 4 = 653251^2 + 83002^2.$$

Taking

$$x = 87, \quad a = 653251, \quad b = 83002, \quad \sigma = 1, \quad \tau = 1$$

gives

$$y = \frac{653251 - 87^3}{2} = \frac{653251 - 658503}{2} = -2626, \quad z = 41501.$$

Hence

$$(87, -2626, 41501)$$

is another integer solution.

6 Concluding form

The equation

$$y^2 + x^3y + z^2 + 1 = 0$$

has no sporadic solutions outside the family described above. Every solution is governed by a representation

$$|x|^6 - 4 = a^2 + b^2$$

with a odd and b even, and the exact admissibility condition on $|x|$ is the two-squares condition

$$v_p(|x|^6 - 4) \equiv 0 \pmod{2} \quad \text{for every prime } p \equiv 3 \pmod{4},$$

together with

$$|x| \equiv 3 \pmod{12}.$$

This gives a full unconditional classification of the integer solutions: not by a finite universal table, but by a complete parametrization and an effective finite procedure for each fixed value of $|x|$.